05D

Alternative Scenario : Skip the Pilot, Go Direct to Production

1,000 OF THE SMALL PARTS, SCALED PROPORTIONALLY

Instead of a small pilot (218 units) followed by a later commercial batch, this scenario assumes moving straight to full commercial production with no field-test phase. The base: **1,000 units of the most-used module (1-gang relay)**, with the other fourteen items scaled at the same original ratio (×25 multiplier).

Product	Original Qty (218)	New Qty (Direct Production)
1-gang relay module	40	1000
2-gang relay module	20	500
4-gang relay module	10	250
3.5″ touch panel	15	375
11″ touch panel	6	150
Smart lock (indoor)	8	200
Smart lock (outdoor)	8	200
Camera (indoor)	15	375
Camera (outdoor)	10	250
Motion sensor	25	625
Light sensor	15	375
Curtain motor	15	375
Water leak sensor	15	375
Gas leak sensor	8	200
Central Hub/Gateway	8	200
Total	**218 units**	**5,450 units**

Key difference from the earlier 1,000-unit scenario: there, 1,000 was the **total** across all items combined. Here, 1,000 is the quantity of **one item only** (the most-used), with every other item scaled proportionally : producing a much larger total batch (≈5,450 units) suited to a full commercial entry from day one.

05D Estimated Manufacturing Cost : Direct Production

FOB CHINA

Product	Qty	Unit Price (USD)	Total (USD)
1-gang relay module	1000	$4 – $7	$4,000 – $7,000
2-gang relay module	500	$6 – $10	$3,000 – $5,000
4-gang relay module	250	$12 – $18	$3,000 – $4,500
3.5″ touch panel	375	$18 – $28	$6,750 – $10,500
11″ touch panel	150	$60 – $90	$9,000 – $13,500
Smart lock (indoor)	200	$35 – $55	$7,000 – $11,000
Smart lock (outdoor)	200	$55 – $90	$11,000 – $18,000
Camera (indoor)	375	$12 – $20	$4,500 – $7,500
Camera (outdoor)	250	$18 – $30	$4,500 – $7,500
Motion sensor	625	$3 – $6	$1,875 – $3,750
Light sensor	375	$3 – $6	$1,125 – $2,250
Curtain motor	375	$15 – $25	$5,625 – $9,375
Water leak sensor	375	$3 – $5	$1,125 – $1,875
Gas leak sensor	200	$8 – $14	$1,600 – $2,800
Central Hub/Gateway	200	$15 – $25	$3,000 – $5,000
Total Manufacturing (FOB)			**$67,100 – $109,550**

Additional Costs (Shipping, Customs, Certification, Identity)

Item	Low (USD)	High (USD)	Note
Shipping (near full-container)	$2,500	$4,500	Better volume economics than small batches
Customs + VAT (≈26% of FOB)	$17,446	$28,483	Proportional to FOB value
SYLTRA identity customization (one-time)	$800	$3,300	First and only batch where this fee applies
SABER + CITC + admin certification (14 models, fixed)	$5,150	$11,700	Unchanged despite the much larger batch

Key insight: regulatory certification cost stays **completely fixed** ($5,150–$11,700) whether the batch is 218 or 5,450 units : making per-unit certification cost here the lowest of any scenario in this plan.

05D Financial Summary : Direct Production

COST, REVENUE, AND MARGIN

Item	Low (USD)	High (USD)
Manufacturing (FOB)	$67,100	$109,550
Shipping	$2,500	$4,500
Customs + VAT	$17,446	$28,483
SYLTRA identity customization	$800	$3,300
SABER + CITC + admin certification	$5,150	$11,700
Estimated Total	**≈ $92,996**	**≈ $157,533**

Potential Revenue at Our Pricing (15% Below Local Market)

Metric	Value
Total potential revenue (SAR)	1,029,625
Total potential revenue (approx. USD)	$274,567
Total landed cost (SAR)	348,735 – 590,749
Estimated gross margin	≈ 43% – 66%

Scenario takeaway: a much larger upfront investment (≈$93K–$158K) compared to the pilot program, with no field-testing phase before full commitment. Expected gross margin stays in a similar healthy range to the other scenarios, but risk is higher since there is no prior proof of concept (no live pilot homes, no real customer feedback) before locking in the full quantity and capital.

05D Comparing the Three Scenarios

WHICH ONE FITS?

Metric	01 : Pilot (218)	02 : Scaled Batch (1,000 total)	03 : Direct Production (5,450)
Estimated initial investment	\$9,600 – \$21,100	\$22,250 – \$41,900	\$93,000 – \$158,000
Field testing before full commitment	Yes (5–8 homes)	Partial (after a successful pilot)	No
Risk level	Low	Medium	High
Certification cost per unit	≈\$22–\$48	≈\$4.7–\$10.7	≈\$0.9–\$2.1 (lowest)
Speed to full commercial market	Slow (staged)	Medium	Fast (immediate)

Recommendation: direct production offers the best per-unit cost efficiency, but it removes the most important investment review checkpoint (pilot success) before committing the largest share of capital. Recommended only if confirmed B2B commitments already exist (contracts with contractors or developers) that justify skipping the field-testing phase.